

EXERCISE 2
SIXOLISE KLAAS
DATA ANALYTICS (AUG)
07 AUG 2026

Question 1: List all distinct departments in the students table.

```
10 --Exercise 2
11 --Question 1: List all distinct departments in the students table.
12
13 SELECT DISTINCT department
14 FROM brightlearn.students.exercise2
15
16 --Question 2: Get the average age of students per department.
```

Add parameter

Table ▾ +

	^A _C department
1	IT
2	HR
3	Finance

Question 2: Get the average age of students per department.

```

16  --Question 2: Get the average age of students per department.
17  SELECT department, AVG(age) AS avg_age
18  FROM brightlearn.students.exercise2
19  GROUP BY department
20
21  --Question 3: Show departments with more than 1 student.
22
23  SELECT department, COUNT(*) AS student_count
24  FROM brightlearn.students.exercise2
25  GROUP BY department

```

Add parameter

Table ▾ +

	A ^B C department	1.2 avg_age
1	IT	20.5
2	HR	22
3	Finance	23

Question 3: Show departments with more than 1 student.

```

21  --Question 3: Show departments with more than 1 student.
22
23  SELECT department, COUNT(*) AS student_count
24  FROM brightlearn.students.exercise2
25  GROUP BY department
26  HAVING student_count > 1
27

```

Add parameter

Table ▾ +

	A ^B C department	1 ² 3 student_count
1	IT	2
2	HR	2

Question 4: Get all students whose age is between 21 and 23

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28 --Question 4: Get all students whose age is between 21 and 23
29 SELECT *
30 FROM brightlearn.students.exercise2
31 WHERE age BETWEEN 21 AND 23
32
33 --Question 5: List all students in the IT or HR department who are older than 21
34 SELECT *
35 FROM brightlearn.students.exercise2

```

Add parameter

Table ▼ +

	¹ ₃ student_id	^A _C name	¹ ₃ age	^A _C department
1	2	Bob	22	HR
2	3	Charlie	21	IT
3	4	Diana	23	Finance
4	5	Eve	22	HR

Question 5: List all students in the IT or HR department who are older than 21.

```

33 --Question 5: `List all students in the IT or HR department who are older than 21`
34 SELECT *
35 FROM brightlearn.students.exercise2
36 WHERE department = 'IT' or department = 'HR' and age > 21
37
38 --Question 6: Show total credits per department, only for departments with more than 5 total credits.
39 SELECT department, SUM(credits) AS total_credits
40 FROM brightlearn.students.courses

```

Add parameter

Table ▼ +

	¹ ₃ student_id	^A _C name	¹ ₃ age	^A _C department
1	2	Bob	22	HR
2	3	Charlie	21	IT
3	4	Diana	23	Finance
4	5	Eve	22	HR

Question 6: Show total credits per department, only for departments with more than 5 total credits.

```

39 SELECT department, SUM(credits) AS total_credits
40 FROM brightlearn.students.courses
41 GROUP BY department
42 HAVING total_credits > 5
43
44 --Question 7: List all courses that do not have 4 credits
45 SELECT *

```

Add parameter

Table ▾ +

	department	total_credits
1	IT	11

Question 7: List all courses that do not have 4 credits

```

44 --Question 7: List all courses that do not have 4 credits
45 SELECT *
46 FROM brightlearn.students.courses
47 WHERE credits != 4
48
49 --Question 8: List all courses that do not have 4 credits
50 SELECT course_id, course_name, credits
51 FROM brightlearn.students.courses
52 SORT BY credits DESC
53

```

Add parameter

Table ▾ +

	course_id	course_name	department	credits
1	101	SQL Basics	IT	3
2	104	Excel	Finance	2
3	105	Statistics	HR	3

Question 8: List all courses that do not have 4 credits

```

49  --Question 8: List all courses that do not have 4 credits
50  SELECT course_id, course_name, credits
51  FROM brightlearn.students.courses
52  SORT BY credits DESC
53  LIMIT 3
54
55  --Question 9: Get the maximum, minimum, and average grade across all enrollments.

```

Add parameter

Table ▾ +

	1 ² ₃ course_id	A ^B _C course_name	1 ² ₃ credits
1	102	Python	4
2	103	Data Science	4
3	101	SQL Basics	3

Question 9: Get the maximum, minimum, and average grade across all enrollments.

```

55  --Question 9: Get the maximum, minimum, and average grade across all enrollments.
56
57  SELECT MAX(grade) as max_grade,
58         MIN(grade) as min_grade,
59         AVG(grade) as avg_grade
60  FROM brightlearn.students.enrollments
61
62  --Question 10
63
64  SELECT course_id, COUNT(course_id) as enrollment_count
65  FROM brightlearn.students.enrollments

```

Add parameter

Table ▾ +

	1 ² ₃ max_grade	1 ² ₃ min_grade	1.2 avg_grade
1	90	78	84.6

Question 10: Count how many enrollments exist per course.

```

64 SELECT course_id, COUNT(course_id) as enrollment_count
65 FROM brightlearn.students.enrollments
66 GROUP BY course_id
67
68 --Question 11: Find total salary and total bonus per department
69

```

Add parameter

Table  

	¹ ₃ course_id	¹ ₃ enrollment_count	
1	101	1	
2	102	1	
3	103	1	
4	104	1	
5	105	1	

Question 11: Find total salary and total bonus per department

```

68 --Question 11: Find total salary and total bonus per department
69
70 Select department, SUM(salary) as total_salary, SUM(bonus) as total_bonus
71 FROM brightlearn.default.salaries
72 GROUP BY department
73
74 --Question 12: Show departments where average salary is above 55,000.
75 SELECT department, AVG(salary) as avg_salary

```

Add parameter

Table  

	^A _C department	¹ ₃ total_salary	¹ ₃ total_bonus	
1	IT	122000	10500	
2	HR	109000	7500	
3	Finance	70000	6000	

Question 12: Show departments where average salary is above 55,000.

```
74 --Question 12: Show departments where average salary is above 55,000.
75 SELECT department, AVG(salary) as avg_salary
76 FROM brightlearn.default.salaries
77 GROUP BY department
78 HAVING avg_salary > 55000
79
80 --Question 13: List employees whose salary plus bonus is greater than 60,000
81 SELECT employee_id,
82        name,
```

Add parameter

Table ▾ +

	^A _C department	1.2 avg_salary
1	IT	61000
2	Finance	70000

Question 13: List employees whose salary plus bonus is greater than 60,000

```
80 --Question 13: List employees whose salary plus bonus is greater than 60,000
81 SELECT employee_id,
82        name,
83        department,
84        salary,
85        bonus,
86        salary + bonus as total_compensation
87 FROM brightlearn.default.salaries
88 WHERE salary + bonus > 60000
89
```

Add parameter

Table ▾ +

	¹ ₃ employee_id	^A _C name	^A _C department	¹ ₃ salary	¹ ₃ bonus	¹ ₃ total_compensation
1	1	Tom	IT	60000	5000	65000
2	3	Spike	Finance	70000	6000	76000
3	4	Tyke	IT	62000	5500	67500